

EXPERIENCE

BSUIR | Full-Stack Software Engineer | Belarus, Minsk (Feb 2025 - Now, 1 year 6 months)

[BSUIR](#) is a leading university in Belarus that graduates specialists in the fields of information technologies and computer science.

- Developed new modules, supported and optimized API applications existing functionality
 - Developed Google Calendar integration for teachers, departments and study groups
 - Optimized individual modules by using database denormalization and migrating to CQRS pattern
 - Implemented a message deduplication mechanism for RabbitMQ message broker
 - **Stack:** TypeScript, NestJS, PostgreSQL, Redis, RabbitMQ, TypeORM
- Improved interface responsibility and UI performance, migrated web applications to a new stack
 - Improved UI performance for large lists and tables by implementing UI virtualization
 - Improved performance and responsibility for complex widgets such as tree lists and lists with multiple filters
 - Simplified api requests by implementing custom HTTP client, generated from Swagger documentation
 - **Stack:** TypeScript, React, Redux Toolkit, Tanstack Query, Bootstrap, React Hook Form
- Improved CI-pipelines execution time and code quality, reduced the technical debt of the project
 - Migrate linting & formatting from ESLint + Prettier stack to BiomeJS, decreasing pipeline execution time from ~35 s to ~6s
 - Migrate testing to Bun Test Runner, decreasing pipeline execution time from by 7 (unit-test) and 1.5 (integration) times
 - Added Knip utility to project which reduced the size of the codebase (about 18K lines of code where deleted)
 - **Stack:** TypeScript, React, Bun test Runner, Testing, BiomeJS, Knip

ltransition | Trainee Full-Stack Engineer | Remotely (Sep 2024 - Nov 2024, Completed)

[ltransition](#) is an international full-service IT integrator company. I got my internship certificate [here](#).

- Developed a presentation editor with real-time multi-user editing and the ability to export presentations to .ppt and .pdf
- Developed an application for real-time messaging with the ability to transfer files and video calls
- Developed a form-building platform for creating forms with the possibility of multi-user mode and analysis of answers
- **Stack:** TypeScript, NestJS, PostgreSQL, TypeORM, React, TailwindCSS, Redux Toolkit + RTK Query, Firebase

OPEN SOURCE

nestjs-multi-limiter | [Github](#) | [npm](#)

A protocol-agnostic, race-condition safe rate limiting module for the [NestJS](#) framework ([Node.js](#)) with 5 core algorithms support.

- Improved Redis storage by executing custom Lua scripts, eliminating race conditions across multi-instance deployments
- Engineered a resilient fault-tolerance mechanism for Redis outages, offering Fail-Open, Fail-Close, and Fail-Fast fallback strategies
- Designed a dynamic options factories mechanism enabling runtime rate-limit configuration on the fly (per request context)
- Architected a protocol-agnostic layer supporting custom key providers and error factories across diverse execution contexts
- Developed a driver-agnostic Redis storage layer completely decoupled from specific underlying npm clients (such as ioredis)
- Implemented native integration with the [@nestjs/swagger](#), automating the generation of customizable API response metadata
- **Stack:** TypeScript, NestJS, reflect-metadata, Redis, [@nestjs/testing](#), Bun Test Runner, GitHub Actions, Turborepo, Biome

EDUCATION

Bachelor of Computing machines, systems and networks | [BSUIR](#) (Minsk, Sep 2021 - Jun 2025, Completed)

Education in the field of computing systems and networks has laid the foundation for a deep understanding of the architecture of modern web applications and the principles of distributed systems.